EGOR SEREBRIAKOV

APPLIED MACHINE LEARNING · DATA SCIENCE

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Summary

M.Sc. student in AI and researcher at Mila with a strong background in **data science, machine learning, optimization, and automation**. Trained through scientific research in Physics and AI, I combine a rigorous analytical mindset and a practical approach to real-world problems. I have hands-on experience in:

(i) **Data analysis and validation:** Python-based analysis to assess data quality, consistency, and reliability.

(ii) **End-to-end data workflows:** building and maintaining data-driven solutions, from requirements definition and data preparation to validation, deployment, and ongoing monitoring.

(iii) **Applied modern machine learning:** from classical ML and deep models for prediction, classification, and anomaly detection to LLMs and agentic workflows, with a focus on validation, robustness, and efficiency.

Skills

Programming	Python *(PyTorch, NumPy, pandas, scikit-learn, OOP)*, SQL
Data Analysis	Exploratory Data Analysis (EDA), Data Validation, Statistical Analysis, Feature Engineering
Machine Learning	LLMs, Agentic AI, Supervised & Unsupervised Learning, Model Validation, Evaluation Metrics, Robustness, Optimization
Tools	Git, Linux, Bash, Jupyter, VS Code
Soft Skills	Communication, Teamwork, Adaptability, Responsibility, Emotional Intelligence

Education

ITMO University
St. Petersburg, Russia (Remote)

M.SC. STUDENT IN ARTIFICIAL INTELLIGENCE, GPA 4.00 — *Sep 2025 - Present*

- Research focus: AI scaling, LLM generalization, time-series modeling, and autonomous AI systems.
- Selected Coursework: *ML Systems Design, Deep Generative Models, Computer Vision, Multimodal LLMs (MLLMs).*

Brown University
Providence, RI

M.SC. IN PHYSICS — *Sep 2022 - May 2024*

- Research focus: AI for particle physics – projects on inference acceleration and anomaly detection.
- Selected Coursework: *Deep Learning, Applied Machine Learning and AI.*

Moscow State University
Moscow, Russia

B.SC. IN PHYSICS WITH HONORS, GPA 3.92 — *Sep 2018 - May 2022*

Projects & Experience

Mila, Québec AI Institute
Montréal, QC

RESEARCH COLLABORATOR – *LLMs & Agentic Systems* — *Mar 2026 - Present*

- Investigating **LLM generalization mechanisms in time-series** through gradient alignment analysis, examining how gradient structure across data distributions relates to transfer behavior.
- Studying **autonomous AI agents**, including orchestration frameworks and AI-scientist architectures, with emphasis on tool use, self-evaluation, monitoring, and safe deployment of autonomous workflows.

ITMO University – Optimization of Physics-Informed Neural Networks
St. Petersburg, Russia (Remote)

RESEARCH ASSISTANT – *Data Analysis / Applied ML* — *Sep 2025 - Apr 2026*

- Developed reinforcement learning–based methods to **automate and optimize training dynamics** of PINNs in nonlinear systems and improved convergence stability through adaptive hyperparameter control.
- Reduced the need for **repeated training sweeps** by showing that a single RL-driven run matched manually tuned causal weights on the Allen–Cahn equation.

Brown University – Transfer Learning for Jet Tagging in Particle Physics
Providence, RI

RESEARCH ASSISTANT – *Data Analysis / Applied ML* — *Jan 2024 - Jun 2024*

- Investigated **knowledge distillation** by transferring information from a large-scale graph neural network (GNN) teacher model to student models trained in two settings: with reduced input resolution and with simplified architecture.
- Achieved **>3x inference speedup** through teacher-student distillation while **preserving classification performance (AUC ≈ 0.9)**, enabling faster jet tagging with the ultimate goal of developing models suitable for real-time deployment at CERN.

Brown University – Anomaly Detection at the Large Hadron Collider
Providence, RI

Research Assistant – *Data Analysis / Applied ML* — *Jan 2024 - Jun 2024*

- Designed and implemented **anomaly detection pipelines** for high-dimensional particle physics data.
- In controlled synthetic stress scenarios, detected **large anomalous regimes** (up to **~50% of samples**), and used these results to analyze model robustness, failure modes, and false-positive risk.

Professional Development

AI Talent Hackathon 2025
Online

Backend & ML Developer — *Sep 2025*

- Developed backend services and data-processing pipelines for an education platform, integrating a language model for automated grading.
- Designed a rubric-based feedback system that generates structured, personalized feedback for each user.
- Implemented a chat assistant and engagement mechanisms (points, badges), and evaluated their impact during live demonstrations.

Princeton University
Princeton, NJ

CERN Computational & Data Science Summer School Student — *Jul 2024*

- Completed hands-on training in parallel programming, optimization, and collaborative software development.
- Worked with CERN and Princeton University researchers on data science and machine learning techniques for high energy physics.

Brown University
Providence, RI

Teaching Assistant — *Sep 2022 - May 2024*

- Teaching Assistant for four physics courses: mentored students during lab sessions, graded coursework, and led problem-solving workshops.

Selected Awards

2025	**Best EdTech Project**, AI Talent Hackathon 2025	*Online*
2025	**Winner**, Junior ML-Contest 2025	*St. Petersburg, Russia*
2022	**Diploma with Honors**, Moscow State University (Top 7%)	*Moscow, Russia*
2019	**Academic Excellence Scholarship**, Moscow State University	*Moscow, Russia*

Presentation

Jan 2026	**AI Talent Conf 2026**, Reinforcement Learning for Physics-Informed Neural Networks	*Online*
Jun 2025	**ML-Contest 2025**, Transfer Learning / Knowledge Distillation for Jet Tagging	*St. Petersburg, Russia*
May 2024	**Deep Learning Day @ Brown**, Machine Learning for Particle Physics	*Providence, RI*

Extracurricular Activity

MSU Academic Group No. 15
Moscow, Russia

Elected Representative — *Sep 2018 - May 2022*

- Represented and protected the interests of a group of 24 students; upheld student rights, ensuring a respectful and inclusive school environment.
- Organized and coordinated student–faculty communication and logistics at the beginning of the pandemic.

Beloretsk Center of Extracurricular Activities
Beloretsk, Russia

Volunteer Teacher — *Oct 2017 - Jun 2018*

- Instructed high school students in mathematics and physics over two semesters.